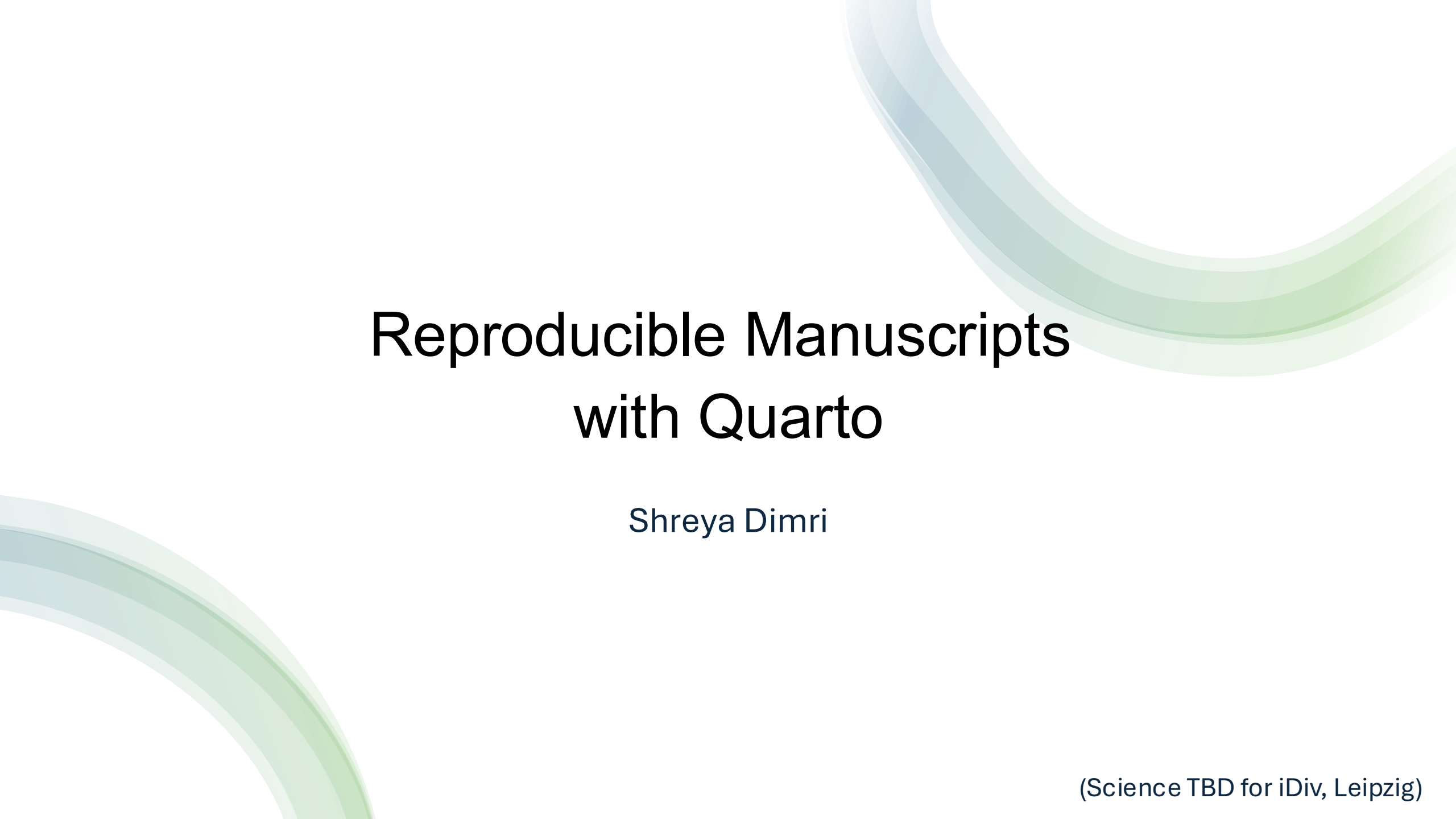




Reproducible Manuscripts with Quarto

Shreya Dimri

WHO AM I?



- Shreya Dimri (she/her)
- PhD Student
- Doing a lot of meta-analysis
- Behaviour ecology and evolution
- Open Science practitioner
- Meta-research enthusiast



<https://github.com/shreyadimri>



<https://bsky.app/profile/shreyadimri.bsky.social>



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But your research is often more than a PDF, it is a “knowledge stack”



Scriberia 

How do we write manuscripts at the moment?

- What do you use?
- And why?

What do we want?

Dynamic Document Generation, where we can:

- combine narrative with code
- automatically generate figures and tables
- automatically render results in text
- format the content into a scientific paper (including citations!)
- Repeat it whenever needed!

Benefits of a dynamic document

Eliminate human error in copying and pasting results

“We found that half of all published psychology papers that use null-hypothesis significance testing (NHST) contained at least one p-value that was inconsistent with its test statistic and degrees of freedom. One in eight papers contained a grossly inconsistent p-value that may have affected the statistical conclusion”

Benefits of a dynamic document

Easy revisions and specification of desired figures and tables

Promote computational reproducibility

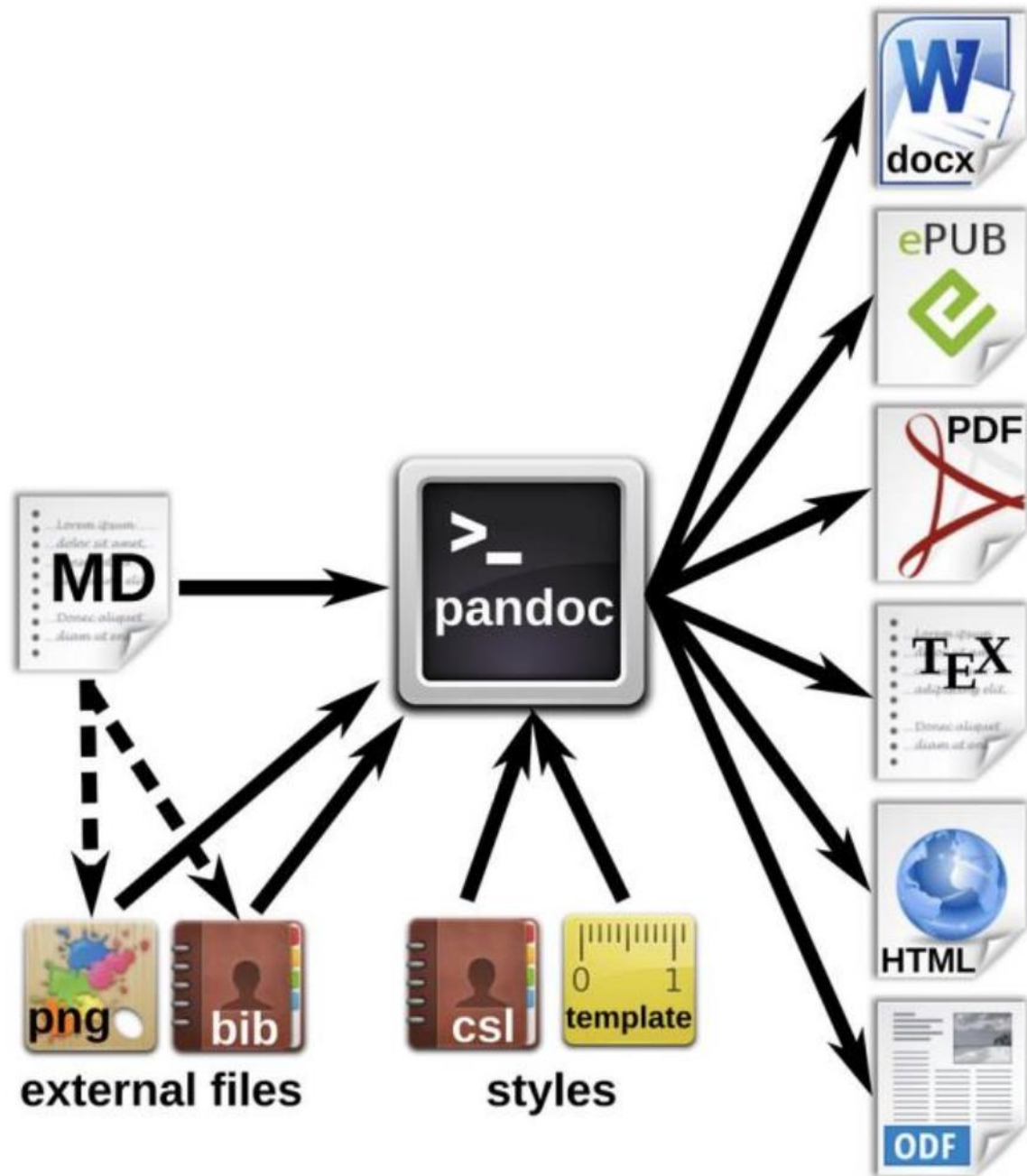
- Easy verification and replication of research findings
- Ultimately reduce mistakes and maybe even save time (once you climb the mount learning)

What is your workflow so far?

- At which stage does reproducibility usually break for you in your analysis?
- Have you ever tried to rerun an old analysis and failed? What was missing?
- What is the hardest part of keeping code, results, and manuscript text in sync?
- What do you currently do manually that you wish were automated?

What is Quarto?

- **Open-source** publishing system (successor to R Markdown)
- Allows for literate programming in Markdown (and with integrated tools and best practices, perhaps making reproducibility a default)
- Versatility (Choice of editors and output formats)
- Supports R, Python, Julia, ObservableJS.. and potentially more in the future
- Exports to HTML, PDF, Word, books, slides, websites.. over 40 possible formats
- Reference management and Cross referencing is possible



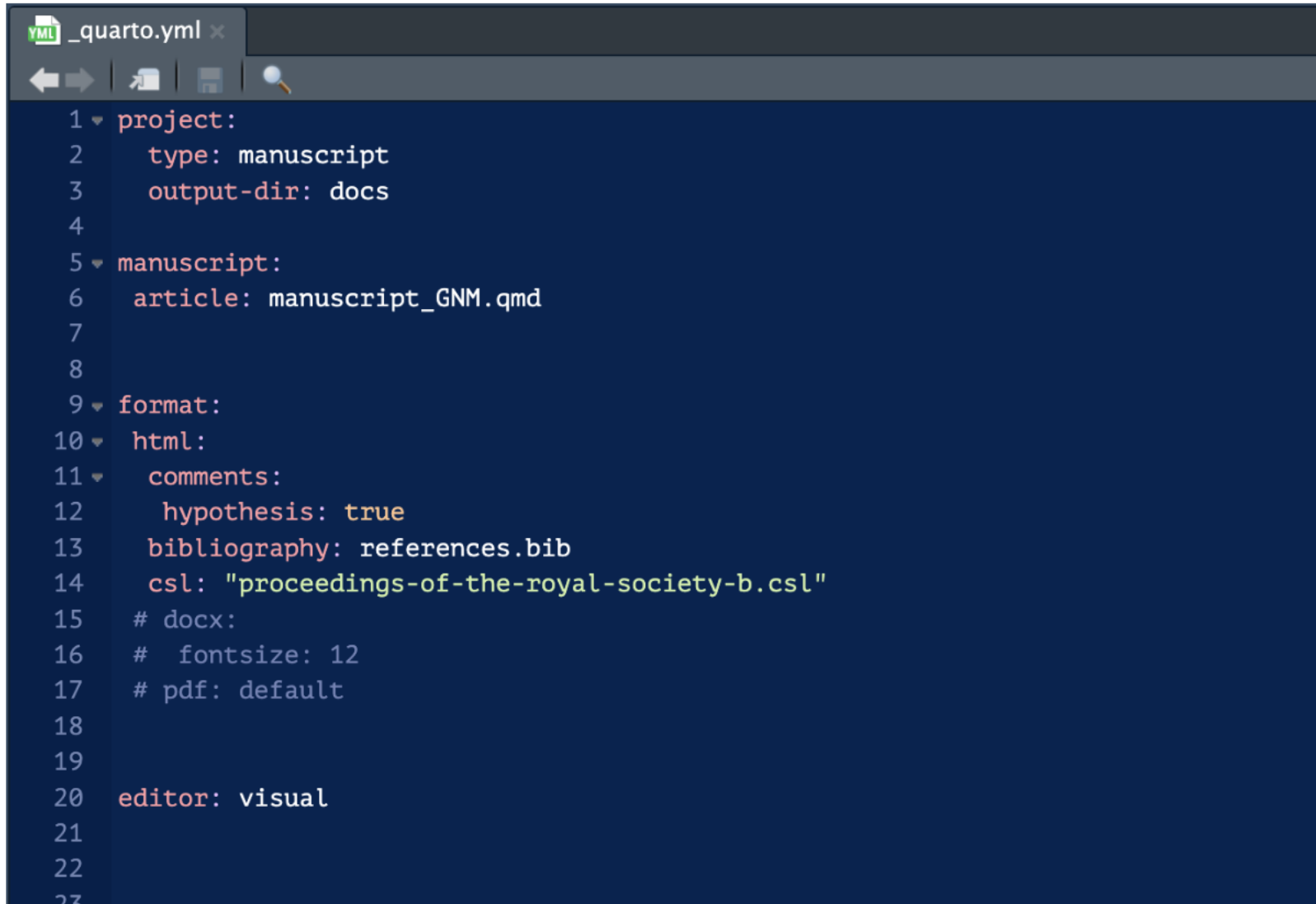
Authoring software options:

- RStudio
- JupyterLab
- VS Code

Components:

- YAML header (metadata)
- Markdown text
- Code chunks + inline code
- Bibliography

YAML header (metadata)



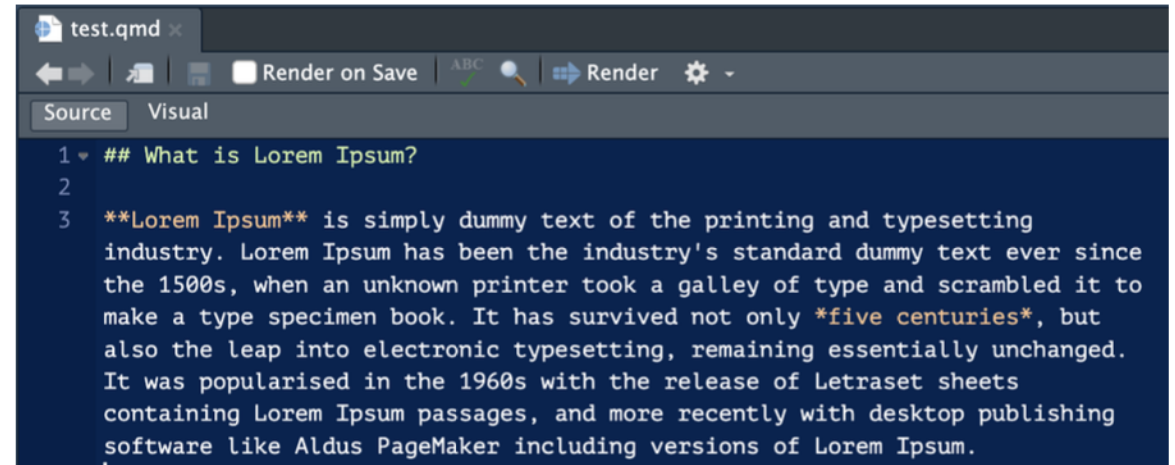
The image shows a code editor window with a dark blue background. The title bar at the top indicates the file is named `_quarto.yml`. The editor contains a YAML configuration for a Quarto project. The configuration is structured as follows:

```
1 project:
2   type: manuscript
3   output-dir: docs
4
5 manuscript:
6   article: manuscript_GNM.qmd
7
8
9 format:
10  html:
11    comments:
12      hypothesis: true
13      bibliography: references.bib
14      csl: "proceedings-of-the-royal-society-b.csl"
15  # docx:
16  #   fontsize: 12
17  # pdf: default
18
19
20 editor: visual
21
22
23
```

Markdown Text

What is Lorem Ipsum?

Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.



```
test.qmd
Render on Save
Render
Source Visual
1 ## What is Lorem Ipsum?
2
3 Lorem Ipsum is simply dummy text of the printing and typesetting
  industry. Lorem Ipsum has been the industry's standard dummy text ever since
  the 1500s, when an unknown printer took a galley of type and scrambled it to
  make a type specimen book. It has survived not only five centuries, but
  also the leap into electronic typesetting, remaining essentially unchanged.
  It was popularised in the 1960s with the release of Letraset sheets
  containing Lorem Ipsum passages, and more recently with desktop publishing
  software like Aldus PageMaker including versions of Lorem Ipsum.
```

References/Bibliography

A typical BibTex key might look like as follows:

```
@article{10.1371/journal.pone.0300333,  
  doi = {10.1371/journal.pone.0300333},  
  author = {Kambouris, Steven AND Wilkinson, David P. AND Smith, Eden T. AND  
Fidler, Fiona},  
  journal = {PLOS ONE},  
  publisher = {Public Library of Science},  
  title = {Computationally reproducing results from meta-analyses in ecology and  
evolutionary biology using shared code and data},  
  year = {2024},  
  month = {03},  
  volume = {19},  
  url = {https://doi.org/10.1371/journal.pone.0300333},  
  pages = {1-22},  
  number = {3},  
}
```

Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.

The screenshot shows the PLOS One website interface. At the top is the PLOS One logo and navigation links: Publish, About, Browse, and a search bar with a magnifying glass icon and the text 'advanced search'. Below the header, there are two icons: 'OPEN ACCESS' and 'PEER-REVIEWED', followed by the text 'RESEARCH ARTICLE'. The main title of the article is 'Computationally reproducing results from meta-analyses in ecology and evolutionary biology using shared code and data'. Below the title, the authors are listed: Steven Kambouris, David P. Wilkinson, Eden T. Smith, and Fiona Fidler. The publication date is March 13, 2024, and the DOI is https://doi.org/10.1371/journal.pone.0300333. On the right side of the article title, there are four yellow buttons: 'Save' (9), 'Citation' (9), 'View' (1,522), and 'Share' (11). Below the article title, there is a horizontal navigation bar with tabs: 'Article' (selected), 'Authors', 'Metrics', 'Comments', 'Media Coverage', and 'Peer Review'. Below the 'Article' tab, there is a list of sections: 'Abstract', 'Introduction', 'Materials and methods', 'Results', 'Discussion', 'Conclusion', 'Supporting information', and 'References'. The 'Abstract' section is currently selected. The abstract text reads: 'Many journals in ecology and evolutionary biology encourage or require authors to make their data and code available alongside articles. In this study we investigated how often this data and code could be used together, when both were available, to computationally reproduce results published in articles. We surveyed the data and code sharing practices of 177 meta-analyses published in ecology and evolutionary biology journals published between 2015–17: 60% of articles shared data only, 1% shared code only, and 15% shared both data and code. In each of the articles which had shared both (n = 26), we selected a target result and attempted to reproduce it. Using the shared data and code files, we successfully reproduced the targeted results in 27–73% of the 26 articles, depending on the stringency of the criteria applied for a successful reproduction. The results from this sample of meta-analyses in the 2015–17 literature can provide a benchmark for future meta-research studies gauging the computational reproducibility of published research in ecology and evolutionary biology.' Below the abstract, there is a 'Figures' section. On the right side of the article, there are two buttons: 'Download PDF' and 'Check for updates'. At the bottom right, there is an advertisement for the 'Tenth International Congress on Peer Review and Scientific Publishing'.

PLOS One

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OPEN ACCESS PEER-REVIEWED

RESEARCH ARTICLE

Computationally reproducing results from meta-analyses in ecology and evolutionary biology using shared code and data

Steven Kambouris, David P. Wilkinson, Eden T. Smith, Fiona Fidler

Published: March 13, 2024 • https://doi.org/10.1371/journal.pone.0300333

Article Authors Metrics Comments Media Coverage Peer Review

Abstract

Introduction

Materials and methods

Results

Discussion

Conclusion

Supporting information

References

Reader Comments

Figures

Abstract

Many journals in ecology and evolutionary biology encourage or require authors to make their data and code available alongside articles. In this study we investigated how often this data and code could be used together, when both were available, to computationally reproduce results published in articles. We surveyed the data and code sharing practices of 177 meta-analyses published in ecology and evolutionary biology journals published between 2015–17: 60% of articles shared data only, 1% shared code only, and 15% shared both data and code. In each of the articles which had shared both ($n = 26$), we selected a target result and attempted to reproduce it. Using the shared data and code files, we successfully reproduced the targeted results in 27–73% of the 26 articles, depending on the stringency of the criteria applied for a successful reproduction. The results from this sample of meta-analyses in the 2015–17 literature can provide a benchmark for future meta-research studies gauging the computational reproducibility of published research in ecology and evolutionary biology.

Figures

9 Save 9 Citation

1,522 View 11 Share

Download PDF

Print Share

Check for updates

ADVERTISEMENT

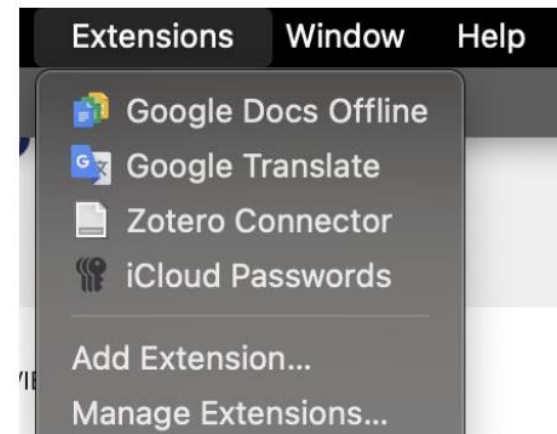
Tenth International Congress on Peer Review and Scientific Publishing

Referencing workflow (example)

My typical workflow is as follows:

- Find a reference that one needs in your manuscript.
- Save it to your Zotero library via the browser extension

Zotero automatically updates your .bib file through syncing with Zotero desktop and updating the .bib file.



Referencing workflow (example)

- Cite the references the @ + reference identifier:
 - **In-text citations:**
 - **Bracketed citations:**

Now you can cite the article in the text you are writing @kambouris2024

Now you can cite the article in the text you are writing [@kambouris2024]

Now you can cite the article in the text you are writing Kambouris et al. (2024)

Now you can cite the article in the text you are writing (Kambouris et al. 2024)

References

Kambouris, Steven, David P. Wilkinson, Eden T. Smith, and Fiona Fidler. 2024.

"Computationally Reproducing Results from Meta-Analyses in Ecology and Evolutionary Biology Using Shared Code and Data." *PLOS ONE* 19 (3): e0300333.

<https://doi.org/10.1371/journal.pone.0300333>.

Limitations of Quarto

- Requires R/Python + (for PDF) LaTeX setup
- Learning curve (Markdown/YAML)
- Less familiar to collaborators/editors
- Journal templates sometimes need manual cleanup
- Smaller community (compared to LaTeX or Rmd)

short-term learning curve for long-term gain

Resources

<https://utrechtuniversity.github.io/workshop-reproducible-manuscripts>

<https://lmu-osc.github.io/introduction-to-Quarto/>

<https://mine.quarto.pub/quarto-manuscripts-rmed/#/whats-next>

<https://quarto.org/docs/manuscripts/>

To the trying out part!